5.1.6 Wrap-Up: Cool Down

1. Solve each equation and check the solution using substitution.

   a. $23.5 - 32p = 16.5p + 4.1$

   b. $\frac{3}{4}(12n - 10) = 19n - 3.5$

Unit 5, Lesson 2: Solving Equations and Rearranging Formulas

Benchmarks

MA.912.AR.2.1 Given a real-world context, write and solve one-variable multi-step linear equations.

MA.912.AR.1.2 Rearrange equations or formulas to isolate a quantity of interest.

Learning Targets

- ☐ I can solve multi-step linear equations.
- ☐ I can isolate a quantity of interest in a formula by rearranging a formula.

5.2.1 Warm-Up: Closest to Zero

1. Place a digit in each blue box to make the equation true for only one value of x. Use only the digits 1 to 9, but digits can be used more than once.

$$\square x + \square = \square x + \square$$

5.2.2 Collaboration: Find the Error

Sandra, Jerry, Marly, and Xavier solved the same equation and each found different solutions. Each student's work and solution are shown in the table.

1. Work with your partner to examine the work shown and determine who solved the equation correctly. Then, for the students who made errors, circle the step where the error was made and explain the error.

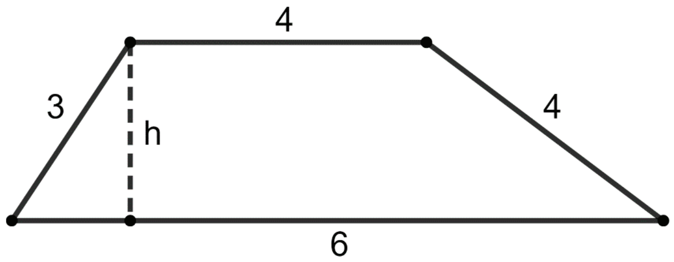
Name	**Student's Work and Solution**	**Is the solution correct?**
Sandra	$9-\frac{3}{4}(16-2w)=\frac{1}{3}w+11$ $\frac{3}{4}(16-2w)=\frac{1}{3}w+2$ $12-\frac{3}{2}w=\frac{1}{3}w+2$ $10-\frac{3}{2}w=\frac{1}{3}w$ $10=\frac{11}{6}w$ $\frac{60}{11}=w$	○ Yes ○ No
	Error Explanation	
Jerry	$9-\frac{3}{4}(16-2w)=\frac{1}{3}w+11$ $9-12+\frac{3}{2}w=\frac{1}{3}w+11$ $-3+\frac{3}{2}w=\frac{1}{3}w+11$ $-3=\frac{7}{6}w+11$ $-14=\frac{7}{6}w$ $-12=w$	○ Yes ○ No
	Error Explanation	

Name	Student's Work and Solution	Is the solution correct?
Marly	$9-\frac{3}{4}(16-2w)=\frac{1}{3}w+11$ $9-12+\frac{3}{2}w=\frac{1}{3}w+11$ $-3+\frac{3}{2}w=\frac{1}{3}w+11$ $\frac{3}{2}w=\frac{1}{3}w+14$ $\frac{7}{6}w=14$ $w=12$	○ Yes ○ No
	Error Explanation	
Xavier	$9-\frac{3}{4}(16-2w)=\frac{1}{3}w+11$ $9-12-\frac{3}{2}w=\frac{1}{3}w+11$ $-3-\frac{3}{2}w=\frac{1}{3}w+11$ $-\frac{3}{2}w=\frac{1}{3}w+14$ $-\frac{5}{6}w=14$ $w=-16.8$	○ Yes ○ No
	Error Explanation	

5.2.3 Guided Instruction: Rearranging Formulas

The formula for the area of a trapezoid is $A=\frac{1}{2}(b_1+b_2)h$, where b_1 and b_2 are the lengths of the bases and h is the height of the trapezoid.

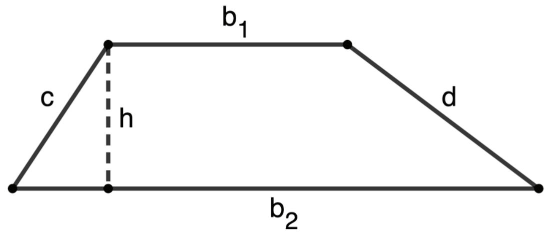
Stacey knows the area of the trapezoid shown is 12.5 square units, and uses the values given to determine the height, h, of the trapezoid. Her work is shown.



1. Complete the table to explain each step of Stacey's work.

Stacey's Work	Explanation of Steps
$A=\frac{1}{2}(b_1+b_2)h$	Formula for area of a trapezoid
$12.5=\frac{1}{2}(4+6)h$	Substitute known values
$25=(4+6)h$	
$\frac{25}{(4+6)}=h$	
$\frac{25}{(10)}=h$	
$2.5=h$	

2. Stacey wants to rearrange the area formula for a trapezoid to isolate the variable h before replacing the values of the other variables.



a. Use Stacey's work on the previous problem to help solve the equation for h.

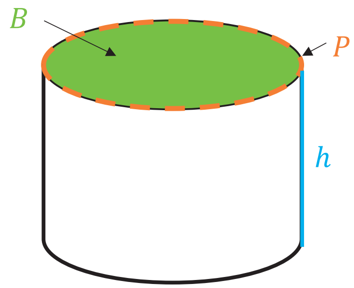
$$A = \frac{1}{2}(b_1 + b_2)h$$

b. Check your work by substituting the values from the previous question. If correct, $h = 2.5$.

c. Explain the similarities and differences between rearranging the formula to isolate h and Stacey's work to find the value of h.

Similarities	**Differences**

3. The formula for the total surface area of a cylinder, S, is $S = 2B + Ph$, where B is the area of the base, P is the perimeter of the base, and h is the height of the cylinder.



Rearrange the formula for the surface area of a cylinder to isolate the variable h.

$$S = 2B + Ph$$

5.2.4 Your Turn!

Rearrange the formulas to isolate the specified variables.

1. The formula for force, F, is mass, m, times acceleration, a, or $F = ma$. Solve for mass, m.

2. The formula $C = 2\pi r$ is used to find the circumference, C, of a circle with radius r. Solve the equation for the radius, r.

3. The formula to convert degrees Fahrenheit, F, into degrees Celsius, C, is $C = \frac{5}{9}(F - 32)$. Solve for F.

5.2.5 Wrap-Up: Growth Mindset

1. Choose **two** of the following prompts to complete.

   a. Describe a new strategy you learned today.

   b. Describe a mistake you made and what you learned from it.

   c. Explain one way you challenged yourself today.